compute_variance_component_colours.R

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`compute_variance_component_colours` *Mix Variance Component Colours*

Description

Mixes component colours per observation using weighted colour-space coordinates.

Usage

```
compute_variance_component_colours(
  data_component_shares,
  vec_component_colours,
  vec_required_components = base::c("Abiotic", "Spatial", "Associations"),
  observation_id_column = "observation_id",
  component_column = "component",
  share_column = "component_share",
  method = base::c("HCL", "perc_avg")
)
```

Arguments

`data_component_shares`
: Data frame with observation identifiers, component IDs, and component shares.

`vec_component_colours`
: Named character vector mapping canonical component IDs to HEX base colours.

`vec_required_components`
: Character vector of required canonical component IDs. Defaults to ‘c("Abiotic", "Spatial", "Associations")‘.

`observation_id_column`
: Single character string naming the observation ID column.

component_column
: Single character string naming the component ID column.

share_column
: Single character string naming the numeric share column.

method
: Character string selecting the colour-mixing method. ‘"HCL"‘ keeps the historical weighted polar LUV/HCL blend. ‘"perc_avg"‘ averages weighted CIE LAB coordinates for a perceptual-average blend.

Value

Tibble with one row per observation and columns ‘observation_id‘ and ‘tile_fill_colour‘.

Examples

```
data_component_shares <-
  tibble::tibble(
    observation_id = base::c(
      "obs_1",
      "obs_1",
      "obs_1",
      "obs_2",
      "obs_2",
      "obs_2"
    ),
    component = base::c(
      "Abiotic",
      "Spatial",
      "Associations",
      "Abiotic",
      "Spatial",
      "Associations"
    ),
    component_share = base::c(50, 30, 20, 30, 30, 40)
  )

compute_variance_component_colours(
  data_component_shares = data_component_shares,
  vec_component_colours = base::c(
    "Abiotic" = "#D95F02",
    "Spatial" = "#7570B3",
    "Associations" = "#1B9E77"
  )
)
```

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